

Rahitha Thangavel

IT Undergraduate

Colombo, Sri Lanka | trahitha10@gmail.com | +94 77 476 1393

 [LinkedIn Profile](#)

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PROFILE

IT undergraduate specializing in Artificial Intelligence with hands-on experience in building LLM-powered applications using Retrieval-Augmented Generation (RAG). Skilled in Python, LangChain, and vector databases to develop intelligent systems for semantic search and question answering. Possesses solid technical knowledge combined with a continuous learning mindset developed through academic projects and practical exposure to modern AI frameworks. Aspires to build a career in Artificial Intelligence and contribute to the development of innovative systems in a dynamic technological environment.

EDUCATION

BSc (Hons) in Information Technology (CGPA: 3.62)

Mar 2024 - Present

Department of Industrial Management | University of Kelaniya

Dean's List — Year 1 (GPA: 3.87)

PROJECTS

Hemas PharmaComply AI – Automated Regulatory Compliance System (Group)

Apr 2026

- Led development of the Impact Predictor module which matches real-time price changes to internal product catalogs and auto-generates actionable business plans
- Integrated and merged teammates' modules (Q&A, deadline tracker, risk evaluator, change detector) into a unified pipeline
- System features automated gazette sync, page-preserving OCR, and a RAG pipeline for citation-backed document Q&A

Tech: Python, LangChain, Groq (Llama 3), ChromaDB, HuggingFace Embeddings, Tesseract OCR, pypdfium2, Streamlit

Machine Learning Project (Group)

Feb 2026 – Present

- Led data preprocessing pipeline — handled missing values, removed duplicates, and prepared behavioral and demographic data for modelling
- Performed feature scaling and feature engineering to improve model accuracy for donor likelihood and donation amount prediction
- Implemented K-Means and Hierarchical clustering for pattern discovery across donor segments
- Team applied classification/regression approaches for final prediction output

Tech: Python, Scikit-learn, Clustering, Classification, Regression

Gazette Chatbot – Retrieval-Augmented Generation (RAG) System (Group)

Jan 2026

- Built the follow-up question handling module — enabling contextually aware multi-turn conversations within the RAG pipeline
- System performs semantic search over Sri Lanka's Legal Gazette documents and returns citation-backed responses
- Integrated with LangChain orchestration, ChromaDB vector store, and Groq (Llama 3) for fast inference

Tech: Python, LangChain, Groq (Llama 3), ChromaDB, HuggingFace Embeddings, Streamlit

CineBook – Distributed Online Movie Seat Booking System (Group)

Jan 2026

- Built the backend concurrency logic — implemented real-time seat locking to eliminate double-booking under simultaneous requests
- Contributed to centralized database management ensuring data consistency across distributed nodes
- Configured NGINX for service routing and load monitoring

Tech: Java, REST APIs, MySQL, NGINX

Software Development Project (Individual)

Jul 2025 - April 2026

Designed and built a full-stack Chess Academy Management System from requirements to development as a year-long individual project.

- Handled end-to-end development: requirements gathering, system analysis, ER diagrams, activity/sequence diagrams, UI design, and full implementation
- Built the backend using Django + Django REST Framework with JWT-based authentication and role-based access for admins, clerks, coaches, and parents
- Developed a React frontend with Bootstrap, enabling management of student registration, attendance, payments, coach salaries, batch scheduling, and tournament registration
- Designed the MySQL database schema and integrated all modules into a cohesive, scalable platform

Tech: React, Django, Django REST Framework, MySQL, JWT, Bootstrap, Figma

SKILLS

AI & LLMs: RAG | LangChain | Llama 3 | Prompt Engineering | Vector Databases (ChromaDB) | HuggingFace Embeddings

Machine Learning: Scikit-learn | Data Preprocessing | Feature Engineering | K-Means Clustering | Hierarchical Clustering | Classification | Regression | Pandas | Numpy

Programming: Python (Strong) | Java | JavaScript | C | C++

Backend APIs: REST APIs | Node.js/Express.js

Databases: MongoDB | MySQL | ChromaDB

DevOps & Cloud: Docker | Microsoft Azure

Soft Skills: Leadership | Adaptability | Problem Solving | Continuous Learning

CERTIFICATIONS

Programming in Python

Sep 2025

Meta

EXTRA CURRICULAR ACTIVITIES

Delegates Coordinator in BR Avenue 25/26

Jul 2025 - Present

LED KLN Club

Oct 2024 - Mar 2025

Member of the Chess Pool

University of Kelaniya

Apr 2025 - Present

Formal Compere/Moderator

Smart Computing and Systems Engineering.(SCSE)

ACHIEVEMENTS

Finalist (Top 10) | MVP Development

Dec 2025

AITHON 2026 | Hemas Holdings

Finalist (Top 5)

Dec 2025

Venture Verse 2025, Sabaragamuwa University

Finalist (Top 10)

Sep 2025

Technopreneur Challenge, CodeFest 2025, SLIIT

REFERENCES

Dr. Thilini Mahanama

Lecturer, Department of Industrial Management

Faculty of Science, University of Kelaniya

Tel : +94 71 760 6243, +94 11 291 4482

E-mail : thilininim@kln.ac.lk

Dinithi Lorensuhewa

Assistant Lecturer

SLIIT Malabe Campus,

Malabe

Tel : +94 77 341 1787

E-mail : dinithi.l@slit.lk